

BT501 MIDTERM SUBJECTIVE

BIOTECH BRAINY BUNCH

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Hemophilia and genetics.

It has been reported that the Arab physician Abu al-Qasim al-Zahrawi, commonly known as Abulcasis, wrote a book on hematology. His book Al-Tasrif written during the period 1000 AD provided the first description of hemophilia, a hereditary genetic disorder

Disorder and Genetics

Hemophilia is a hereditary bleeding disorder resulting from a partial or total lack of an essential blood clotting factor. It is a lifelong disorder that produces excessive bleeding, and spontaneous internal bleeding occurs very frequently. Hemophilia A, the most common form, is caused by a deficiency in clotting factor 8 (factor VIII). Hemophilia B results from a deficiency in clotting factor 9(IX). This disorder is inherited in a sex linked recessive manner.

Parkinson's disease and its conditions.

After Alzheimer disease, Parkinson disease is the second most common neurodegenerative disease. Parkinson disease affects about 1% of the population that is 60 years of age or older and is characterized by the progressive reduced capacity to initiate voluntary movements arising mainly from the loss of neurons that synthesize the neurotransmitter dopamine. Among the many factors that mediate the development of Parkinson disease, mitochondrial dysfunction is considered to be a major factor in its etiology and pathogenesis.

The dysfunction of mitochondria in Parkinson disease patients may result from one or more of the following conditions:

- Deletion of mitochondrial DNA,
- Accumulation of mitochondrial DNA
- Mutations, increase in oxidative stress from reactive oxygen species
- Deficient expression and function of the mitochondrial respiratory chain
- Abnormal morphology of mitochondria.

Neurodegenerative diseases such as Parkinson disease are often closely associated with an imbalance of mitochondrial homeostasis. The levels of many mitochondrial proteins are altered in postmortem samples of brains from persons with Parkinson disease. Genetic mutations in the PTEN-induced putative kinase 1 (PINK1), parkin (PARK2), DJ-1, alpha-synuclein (SNCA), and leucine-rich repeat kinase 2 (LRRK2) genes are closely linked to the recessive Parkinson disease genes. Moreover, these mutations are associated with important functions that maintain mitochondrial homeostasis, which include membrane potential, calcium homeostasis, structure of cristae (internal compartments formed by the inner membrane of mitochondria), respiratory activity, mitochondrial DNA integrity, and clearance of dysfunctional mitochondria from cells.

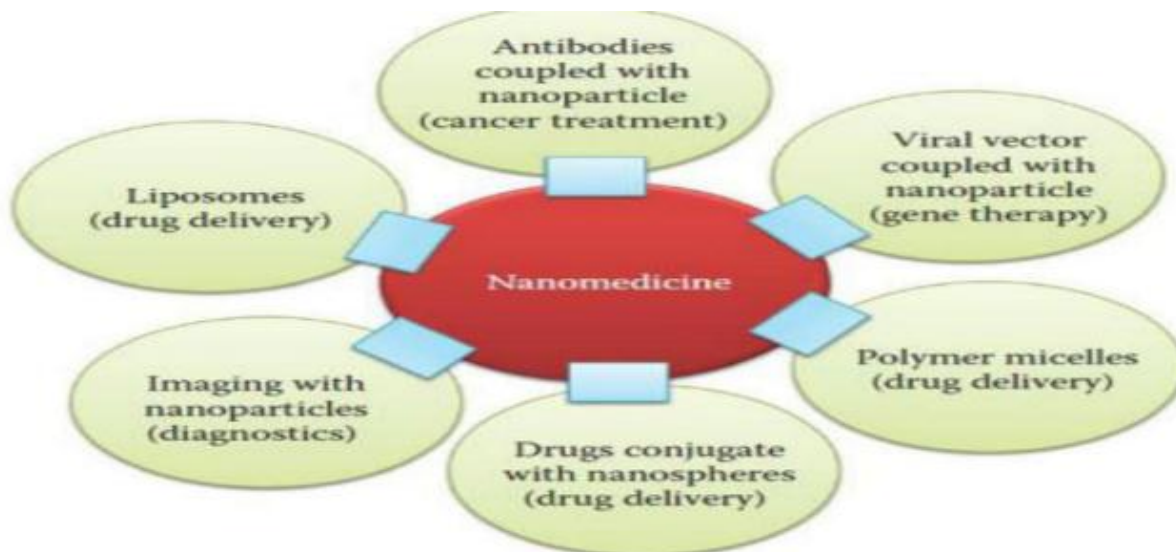
Cardiovascular diseases.

Major cardiovascular diseases include several subtypes, such as coronary heart disease, cerebrovascular disease, heart failure, rheumatic heart disease, and congenital heart disease. Cardiovascular disease is a leading health problem, affecting more than 80 million individuals in the United States. In Canada, 1 out of 3 deaths is attributable to cardiovascular disease.

The major risk factors associated with cardiovascular diseases are cigarette smoking, unhealthy diet, physical inactivity, hypertension, diabetes, and high blood cholesterol. Cardiovascular diseases may also result from a variety of genetic causes, including single gene mutations, interactions between multiple genes, and gene–environment interactions.

Economic shifts, urbanization, industrialization, and globalization bring about lifestyle changes that promote heart disease. In developing countries, life expectancy is increasing rapidly and is expected to expose people to these environmental risk factors for longer periods.

Nanobiotechnology and its applications



Explain RNAi mechanism of Gene expression

The RNA interference (RNAi) is a natural process that cells dictate to turn down or go to a silent state; such activity cell is regulated by a specific gene known as RNAi. The RNAi technology provides excellent tools to study the role specific gene in the development, causes, and progression of particular diseases, and has taken the biomedical community by storm. It has been reported that RNAi was first noticed in the plant *Petunia* when plant biologists tried to develop the purple color of the flower by introducing a pigment-producing gene. Surprisingly, the gene which was introduced to intensify color in the plants was found to suppress the color. In the end the resulting flowers developed white patches or became completely white. Surprisingly, a few years later another group of researchers observed a similar kind of gene-silencing effect in *Caenorhabditis elegans*.

Later on, it has been reported that the gene-silencing effect is caused by the presence of double-stranded RNA, and this double-stranded RNA is normally not found in healthy cells. RNAi technology has been in use in various fields of biotechnology, particularly in the food plant engineering that produces lower levels of natural toxins. No plant that uses RNAi technology has yet passed the experimental stage; however, research work has been known to effectively reduce the allergen levels in tomato plants and also is known to cut the cancer causing agents in tobacco plants. Also, other plant characters that have also been bioengineered include the production of nonnarcotic natural products by using the opium plant, development of resistance to common plant viruses, and protection of plants with dietary antioxidants. Interestingly, bioengineered plants such as *FlavrSavr* tomatoes and two cultivars of ring-spot-resistant papaya were originally developed by employing antisense technology. One of the pronounced applications of RNAi technology is its application in medicine. Over the past few years, efforts have been made to understand RNAi's role in normal and diseased cells, and also to use the RNAi technology for use in medical therapies. It has been suggested that human disease progression can be blocked by using RNAi-based therapies to turn down the activity of genes.

Cancer, for example, is frequently caused by over-excited genes in the cells, and retarding their activities could stop the disease progression. Over the last few years, several pharmaceutical companies are using RNAi-based therapies to treat for various forms of cancer. In addition, viral infections can also be treated using RNAi-based therapies and this can be done by reducing the activity of key viral genes. It has been shown in the laboratory that human cells have successfully stopped the growth of HIV, polio, hepatitis C, and other viruses in human cell culture and RNAi-based therapies against HIV are under clinical trial stages.

Moreover, the importance of RNAi technology has some beneficial effects in finding out the cause of the disease. Using RNAi technology, the activity of a particular gene can be knocked down which will help us to understand its role in the disease development and progression. For many years, researchers have been studying how proteins regulate gene activity. Now with the help of RNAi technology, it would be possible to discover the role of proteins in gene regulation.

Applications of DNA microarray

DNA microarray has increasing number of applications:

1-Expression analysis : In this method, the DNA of an organism is spotted on a chip and hybridized to cDNA obtained from mRNA of the same organism. The intensity signifies the abundance of a particular mRNA in the sample, signifying the expression level of the gene under a given condition. This is the most commonly used technique.

2-Mutation analysis : The genomic DNA is spotted on a chip and hybridized to DNA isolated from different individuals. Single nucleotide polymorphism can be identified by this method. It is important to analyze variation in a particular gene.

3- Comparative genomic hybridization: Increase or decrease of chromosomal fragments associated with a disease state. Prenatal chromosomal aberrations can be studied.

Diagnosis through ELISA for:

- **part.1 hepatitis antigen detection**

Detection of the viral surface antigen with a sandwich ELISA utilizes specific monoclonal antibodies to capture the hepatitis B surface antigens present in blood in wells of an assay plate (Fig. A). The monoclonal antibodies were generated using recombinant hepatitis B surface protein expressed in yeast. An anti-hepatitis B surface antigen detection antibody with a conjugated enzyme and the enzyme substrate are added to measure virus titers.

- **Part 2.detection of hepatitis B virus antibody**

An indirect ELISA that detects the presence of anti-hepatitis B virus antibodies in the blood sample is often used to confirm a positive result; however, blood from individuals immunized with a hepatitis B vaccine also produces a positive result with this test. Blood may additionally be screened for hepatitis B DNA and C virus RNA.

Allele's specific hybridization of cystic fibrosis

In addition to infectious diseases, hybridization probes are widely used to detect specific disease-associated alleles. Monogenic diseases are caused by mutations in a single gene; however, any one of several alterations to the normal nucleotide sequence of a gene may be responsible.

This is exemplified by cystic fibrosis, a common lethal autosomal recessive disorder that affects approximately 1 in every 2,500 live births. Mutations in a single gene, the cystic fibrosis transmembrane conductance regulator (CFTR) gene, result in defects in chloride ion transport.

As a consequence, the mucus of lung and other mucosal tissues is thick and viscous, obstructing the respiratory, digestive, and reproduction system functions. Every state in the United States now routinely screens newborns for cystic fibrosis. An ELISA is used to detect higher-than-normal levels of immunoreactive trypsinogen in blood. This protein is produced by the pancreas

and is linked to cystic fibrosis. Premature babies and babies from stressful births also may have elevated levels of immunoreactive trypsinogen, and in these cases, genetic tests are employed for confirmation.

About 1,900 different mutations are reported to occur in the CFTR genes of patients with cystic fibrosis. Screening individuals who may be at risk for cystic fibrosis for such a large number of possible mutations is a daunting task. However, some of the mutations that cause cystic fibrosis are much more common than others. The most common mutation is an inframe deletion of three nucleotides in exon 10 of the CFTR gene that leads to loss of the amino acid phenylalanine at codon 508 ($\Delta F508$). Over 90% of cystic fibrosis patients carry at least one $\Delta F508$ allele, and nearly 50% of cystic fibrosis patients are individuals who are homozygous for $\Delta F508$. It is estimated that about 160 different mutations account for 96 to 97% of cystic fibrosis alleles.

What is the difference between classical biotechnology and modern biotech?

- Classical Biotech –since ages ,Growing better crops, Breeding animals
- Cheese and yogurt production – use of microorganisms
- Mendel's work – Laying foundation of modern Biotech
- Major discoveries : DNA structure, Ligase, Restriction enzyme
- Birth of rDNA technology
- Cut Gene from one organism and paste into others

Steps of indirect ELISA for RF factor ?

Indirect enzyme-linked immunosorbent assay (ELISA) for Rheumatoid Factor (RF) involves several key steps.

1. **Coating:** Antigen (RF) is immobilized on a microtiter plate.
2. **Blocking:** Non-specific binding sites on the plate are blocked to prevent false positives.
3. **Serum Incubation:** Patient serum, containing antibodies, is added and allowed to bind to the immobilized antigen.
4. **Washing:** Unbound serum components are washed away to reduce background noise.
5. **Secondary Antibody Addition:** A labeled secondary antibody, specific to human immunoglobulins, is added. This binds to any RF antibodies present.
6. **Second Incubation and Washing:** The plate is incubated to allow the secondary antibody to bind, followed by another washing step to remove unbound secondary antibody.
7. **Substrate Addition:** A substrate for the enzyme label is added, generating a measurable signal.

8. **Signal Measurement:** The optical density or fluorescence is measured, providing a quantitative indication of RF antibody presence.
9. **Data Analysis:** Results are compared to a standard curve or control, determining RF levels in the patient's sample.

three types of trisomy

- **Trisomy 21**, the first human chromosomal disorder discovered (in 1959), is an abnormality that displays an extra copy (total of 3 copies) of chromosome 21 (Fig. 2) and causes Down syndrome.
- **Trisomy 13**, the presence of three copies of chromosome 13, causes Patau syndrome.
- **Trisomy 18**, the presence of 3 copies of chromosome 18, elicits Edward syndrome

how many types of structural variants what is importance of structural variants

In analyzing the role of structural gene variants in cell function, it is important to consider the two types of such variants, i.e., gain-of-function variants and loss-of-function variants.

1. A **gain-of-function variant** results from a mutation that confers new or enhanced activity on a protein. Most mutations of this type are not heritable (germ line) but, rather, are somatic mutations. A change in the structure of a gene that may arise during DNA replication and is not inherited from a parent, and also is not passed to offspring, is called a somatic mutation. Such a mutation that results in a single base substitution in DNA is known as a somatic point mutation.
2. A **loss-of-function variant** results from a point mutation that leads to reduced or abolished protein function. Most loss-of-function mutations are recessive, indicating that clinical signs are typically observed only when both chromosomal copies of a gene (one being inherited from each parent) carry such a mutation.

What is recommendation give by organization to ethical and legal gene therapy

Nevertheless, there are some ethical concerns which have been raised in different parts of the world, and some ethical organizations have released recommendations to resolve the gene therapy issues and these recommendations include:

- (a) establishing a national ethics body in each country to look at somatic gene therapy;
- (b) supporting somatic gene therapy research that follows the recommendations
- (c) asking researchers, organizations, and governments to listen and respond to public concerns about gene therapy research
- (d) asking that research follows quality and safety controls.

what is rDNA applications in medicine , agriculture and pharmacology

- **Agriculture:** growing crops of your choice (GM food), pesticide resistant crops, fruits with attractive colors, all being grown in artificial conditions
- **Pharmacology:** artificial insulin production, drug delivery to target sites
- **Medicine:** gene therapy, antiviral therapy, vaccination, synthesizing clotting factors
- **Other uses:** fluorescent fishes, glowing plants etc

Advantages of liposome based drug delivery?

Their advantages are:

- Biocompatibility
- low toxicity
- non- immunogenicity
- improved drug efficacy. T
- they fuse to cell membrane and deliver the drug.
- They are used as carriers of antifungal, antibacterial, and cancer drugs

Genetics of Tay Sachs ?

Tay–Sachs disease is an autosomal recessive fatal genetic disorder caused by a genetic mutation in the HEXA gene on human chromosome 15, which encodes the hexosaminidase A enzyme. This mutation leads to a decrease in function of hexosaminidase A, and as a result, harmful quantities of a sphingoglycolipid, termed ganglioside GM2, accumulate in brain neurons.

HEXA gene mutations are rare and are most frequently detected in genetically isolated populations. Tay–Sachs disease can occur from the inheritance of either two similar or two unrelated mutations in HEXA that cause disease. Many HEXA mutations have been identified, and these mutations can reach significant frequencies in certain populations .

In the most common form of the disease (infantile Tay–Sachs), abnormal hexosaminidase A enzyme activity and the accompanying harmful accumulation of cell membrane-associated gangliosides in neurons lead to premature neuronal death, paralysis, dementia, blindness, psychoses, and even death of the patient. Although the degeneration of the central nervous system begins at the fetal stage, the loss of peripheral vision and motor coordination are not evident until ~6 months of age, and death usually results by 4 years of age.

Genetic of Huntington disease

Huntington disease is an autosomal dominant genetic disorder; if one parent carries the defective Huntington disease gene, his or her offspring have a 50-50 chance of inheriting the disease. Huntington disease is a neurodegenerative brain disorder in which afflicted individuals lose their ability to walk, talk, think, and reason. They easily become depressed, lose their short-term memory, and may experience a lack of concentration and focus. Every individual with the

gene for the disease eventually develops the disease. The Huntington disease gene is located on the long arm of human chromosome 4 and encodes a protein called huntingtin that is quite variable in its structure. The 5' end of the Huntington disease gene contains many repeats of the CAG trinucleotide (encodes glutamine). A highly variable number of CAG trinucleotide repeats accounts for the Huntington disease gene mutation, which leads to the expression of an abnormally long polyglutamine tract at the N terminus of the huntingtin protein beginning at residue 18. Such polyglutamine tracts increase protein aggregation, which may alter cell function. Thus, like fragile X syndrome, Huntington disease is one of 14 trinucleotide repeat disorders that cause neurological dysfunction in humans. More specifically, Huntington disease is classified as a **polyglutamine disorder**.

Healthy unaffected persons have a CAG repeat count of 9 to 35 (Table). However, alleles with more than 36 CAG repeats give rise to Huntington disease (the highest reported repeat length is 250). Incomplete penetrance is found in alleles with 36 to 39 CAG repeats. People with 36 to 40 CAG repeats may or may not develop Huntington disease, while people with more than 40 CAG repeats are rather likely to develop the disorder. Alleles with more than 60 CAG repeats result in a severe form of Huntington disease known as juvenile Huntington disease, and children who get Huntington disease may range from 2 to 20 years of age

The mass of the huntingtin protein depends mainly on the number of its glutamine residues. Wild-type (normal) huntingtin consists of 3,144 amino acids, contains 6 to 35 glutamines, and has a mass of ~350 kilodaltons (kDa). In Huntington disease patients, huntingtin contains more than 36 glutamines and has an overall higher molecular mass than wild-type huntingtin.

Conditions of dysfunction of mitochondrial parkinson disease

Fragile X syndrome

Disorder and Genetics:

Fragile X syndrome is caused by a —fragile site at the end of the long arm of the X chromosome. This syndrome is manifested by many changes in behavior and cognitive recognition that vary widely in severity among patients. Fragile X syndrome is the most common cause of inherited mental retardation. Although it is an X-chromosome-linked recessive trait with variable expression and incomplete penetrance, 30% of all carrier women are affected.

Fragile X syndrome is caused by loss-of-function mutations in the fragile X mental retardation 1 (FMR1) gene. FMR1 encodes the FMRP protein found in many tissues and at particularly high levels in the brain and testes. In the brain, it may play a role in the development of neuronal synapses and cell communication. The synapses can change and adapt over time in response to experience, a characteristic called **synaptic plasticity**. The FMRP protein may help regulate synaptic plasticity and thereby control learning and memory.

Fragile X syndrome belongs to a growing class of neurodegenerative disorders known as trinucleotide repeat disorders. Among these disorders, 14 affect humans and elicit neurological dysfunction. Trinucleotide CGG repeat expansions (200 to more than 1,000 repeats) that inactivate the FMR1 gene are the most common mutations observed at this locus.

How to produce monoclonal antibodies-3 marks

Producing monoclonal antibodies involves several key steps. First, immunize a mouse with the target antigen to stimulate antibody production. Harvest spleen cells containing antibody-producing B cells. Fuse these cells with myeloma cells to create hybridoma cells capable of indefinite division. Screen hybridomas for antibody secretion and select clones producing the desired monoclonal antibody. Culture selected clones and harvest the antibodies. Purify antibodies using techniques like chromatography. Validate antibody specificity and functionality through assays. Scale up production for larger quantities. This process ensures consistent, high-quality monoclonal antibodies for various applications, such as diagnostics and therapeutics.

Explain different types of monogenic disorders-5 marks.

Monogenic diseases can be classified into three main categories: autosomal dominant, autosomal recessive, and X linked. All humans have two chromosomal copies of each gene or allele, one allele per each member of a chromosome pair.

1. **Dominant monogenic disorders** involve a mutation in only one allele of a disease-related gene. In autosomal dominant inheritance, an affected person has at least one affected parent, and affected individuals have a 50% chance of passing the disorder on to their children. Huntington disease is an example of an autosomal dominant disorder.
2. **Recessive monogenic disorders** occur due to a mutation in both alleles of a disease-related gene. In autosomal recessive inheritance, affected children are usually born to unaffected parents. Parents of affected children usually do not have disease symptoms but carry a single copy of the mutated gene. There is an increased incidence of autosomal recessive disorders in families in which parents are related. Children of parents who are both heterozygous for the mutated gene have a 25% chance of inheriting the disorder, and the disorder affects either sex. Cystic fibrosis and sickle-cell anemia are examples of autosomal recessive disorders.
3. **X-linked monogenic disorders** are linked to mutations in genes on the X chromosome. The X-linked alleles can also be dominant or recessive. These alleles are expressed in both men and women, more so in men, as they carry only one copy of the X chromosome (XY), whereas women carry two (XX).

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What is Parkinson's disease?

Parkinson's disease is a brain disorder that causes unintended or uncontrollable movements, such as shaking, stiffness, and difficulty with balance and coordination. Symptoms usually begin gradually and worsen over time. As the disease progresses, people may have difficulty walking and talking.

Define Huntington disease?

Huntington's disease is a rare, inherited disease that causes the progressive breakdown (degeneration) of nerve cells in the brain. Huntington's disease has a wide impact on a person's functional abilities and usually results in movement, thinking (cognitive) and psychiatric disorders. Huntington's disease is caused by a faulty gene that results in parts of the brain becoming gradually damaged over time. Define Basophil? Basophils are a type of white blood cell. Basophils are the least common type of granulocyte, representing about 0.5% to 1% of circulating white blood cells. However, they are the largest type of granulocyte.

Types and names of immune disease?

- Hashimoto thyroiditis
- Celiac disease
- Graves' disease
- Addison's disease
- Inflammatory bowel disease
- Multiple sclerosis
- Rheumatoid arthritis
- Type 1 diabetes

Name of bio marker for ovarian carcinoma

CA125 is the best-known serum ovarian cancer biomarker, it is not the only one: CEA (mucinous), LDH (dysgerminoma, mixed germ cell tumors), β -hCG (choriocarcinoma, mixed germ cell tumors), inhibin B (granulosa cell tumors), α -fetoprotein (yolk sac tumors, embryonal cell tumors), and HE4 are also available.

Steps of sandwich ELISA

- Coat a surface with capture antibody.
- Block unbound protein binding sites on the surface
- Add the sample containing the target antigen

- Add the detection antibody label conjugate
- Detection

Human genome project?

The Human Genome Project (HGP) was an international scientific research project with the goal of determining the base pairs that make up human DNA, and of identifying, mapping and sequencing all of the genes of the human genome from both a physical and a functional standpoint.

Goals of the human genome project include:

1. Optimization of the data analysis.
2. Sequencing the entire genome.
3. Identification of the complete human genome.

What is Immunity?

Immunity is the ability of the body to defend itself against disease-causing organisms. Everyday our body comes in contact with several pathogens, but only a few results into diseases. The reason is, our body has the ability to release antibodies against these pathogens and protects the body against diseases. This defence mechanism is called immunity.

Types of Immunity

There are two major types of immunity:

- Active immunity: Active immunity occurs when our own immune system is responsible for protecting us from a pathogen.
- Passive immunity: Passive immunity occurs when we are protected from a pathogen by immunity gained from someone else.

Difference between innate and adaptive immunity system?

Innate immunity: Innate immunity refers to nonspecific defense mechanisms that come into play immediately or within hours of an antigen's appearance in the body. These mechanisms include physical barriers such as skin, chemicals in the blood, and immune system cells that attack foreign cells in the body. The innate immune response is activated by chemical properties of the antigen.

Adaptive immunity : Adaptive immunity refers to antigen-specific immune response. The adaptive immune response is more complex than the innate. The antigen first must be processed and recognized. Once an antigen has been recognized, the adaptive immune system creates an

army of immune cells specifically designed to attack that antigen. Adaptive immunity also includes a "memory" that makes future responses against a specific antigen more efficient.

What does the red color indicate in DNA microarray?

A red spot indicates that that gene was strongly expressed in cancer cells.

What is metagenomics?

Metagenomics is the study of the structure and function of entire nucleotide sequences isolated and analyzed from all the organisms (typically microbes) in a bulk sample.

Different between orthologs and paralogs?

- orthologs are defined as homologs in different species that catalyze the same reaction.
- Paralogs are defined as homologs in the same species that do not catalyze the same reaction.

Which chromosome has maximum and minimum number of gene?

In the human genome, chromosome 1 has the most genes. It has 2968 genes. Chromosome Y contains the least number of genes.

What is Xist activity regulatory?

Xist (X-inactive specific transcript) is a non-coding RNA on the X chromosome of the placental mammals that acts as a major effector of the X-inactivation process. It is a component of the Xic – X-chromosome inactivation centre along with two other RNA genes and two protein genes.

Structure of the following types of viruses e. g. Single strand or double?

- Adenovirus Double-stranded DNA genome
- Hepatitis B Partially double-stranded DNA
- Polio virus Single-stranded RNA
- Influenza virus Single stranded RNA

What is DNA microarray?

The DNA microarray is a tool used to determine whether the DNA from a particular individual contains a mutation in genes like BRCA1 and BRCA2. The chip consists of a small glass plate encased in plastic.

Note on Cloning Vectors?

Cloning Vector “A cloning vector is a small piece of DNA into which a foreign DNA can be inserted for cloning purposes.” Cloning vectors are utilized to insert foreign DNA into another

cell and create multiple copies of the same. The foreign DNA is duplicated and expressed utilizing the host cell machinery. It amplifies one copy of DNA into multiple copies.

Features of Cloning Vectors

The cloning vectors possess the following features:

- A cloning vector should possess an origin of replication so that it can selfreplicate inside the host cell.
- It should have a restriction site for the insertion of the target DNA.
- It should have a selectable marker with an antibiotic resistance gene that facilitates screening of the recombinant organism.
- It should be small in size so that it can easily integrate into the host cell.
- It should be capable of inserting a large segment of DNA.
- It should possess multiple cloning sites.
- It should be capable of working under the prokaryotic and eukaryotic systems.

Types of Cloning Vectors

There are the following different types of cloning vectors:

- Plasmids
- Bacteriophage
- Phagemids
- Bacterial Artificial Chromosomes

Advantages of PCR-based tests?

- Faster Results
- Shorter Time to Optimal Therapy
- Improve Treatment Decisions
- Avoid Unnecessary Antibiotics
- Reduce Unnecessary Testing
- Reduce Healthcare Costs

What is Real-time PCR or qPCR?

A real-time polymerase chain reaction is a laboratory technique of molecular biology based on the polymerase chain reaction. It monitors the amplification of a targeted DNA molecule during the PCR, not at its end, as in conventional PCR. Real-time PCR can be used quantitatively and semi-quantitatively.

Recombinant DNA technology and its rDNA cloning steps?

Recombinant DNA technology Recombinant DNA technology is the joining together of DNA molecules from two different species. The recombined DNA molecule is inserted into a host organism to produce new genetic combinations that are of value to science, medicine, agriculture, and industry.

Steps of Recombinant DNA technology

Step-1. Isolation of Genetic Material.

Step-2. Cutting the gene at the recognition sites.

Step-3. Amplifying the gene copies through Polymerase chain reaction (PCR).

Step-4. Ligation of DNA Molecules.

Step-5. Insertion of Recombinant DNA Into Host.

Point independent method of mapping for diseased gene?

In Gene mapping Recombination frequency and gene distance in the chromosome are directly proportional. On one chromosome, there is a different genes present at a different distance. So there is a direct relation between a gene distance and a recombination frequency. For disease it is important and measured by the linear order by genetic mapping.

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